Strategic Analysis of BrightTV : Methods and Insights

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Objective

Goal: Increase BrightTv's subscription base within this financial year.

CEO's request: Insights for CVM (Customer Value Management) team to achieve the objective.

Data Overview

- The dataset attached contains information on the user profiles and viewers transactions for BrightTv.
- The dataset was from the months January, February and March from the year 2016.
- Times and dates in the dataset are supplied in UTC.
- Consumption is split per session, i.e. for every session a subscriber has there will be 1 record.

Tools and Techniques

- Data Analytics Tools: SQL on Snowflake and MS Excel for statistical analysis and visualization.
- Visualization: PowerPoint for presentation and Power BI for Dashboard.

Understanding the dataset

1. Joining the two tables User Profile and Viewership to make analysing the data easier.

```
--JOIN THE TWO TABLES

SELECT * FROM userprofile AS u
JOIN viewership AS v ON u.userid = v."UserID";

```

```
--Joined table named tv

SELECT * FROM broadcast.public.tv
LIMIT 10;

```

2. The table tv has 3 columns names userid, "userid" and "UserId" our objective would then be to check if these columns contain the same in formation using:

```sql
--checking if the userid columns are identical

SELECT userid,"userid", "UserID"
FROM broadcast.public.tv
LIMIT 10;
```

This SQL script shows that the three columns are identical resulting in us removing the two extra columns using:

```sql
--removing the two columns userid as they are three identical columns

ALTER TABLE broadcast.public.tv
DROP COLUMN "userid", "UserID";
```

The extra columns are being dropped to avoid confusion as there already is a column named userid.

Please note that the reason userid is in parenthesis is due to the auto fix that occurred while uploading the data into snowflake, the system automatically renamed the two columns with parentheses.

3. Checking for NULL values
   Script used to identify NULL/Missing values

```sql
--Checking for Nulls

SELECT
     SUM( CASE WHEN userid IS NULL THEN 1 ELSE 0 END) AS userid_id_missing,
     SUM( CASE WHEN name IS NULL THEN 1 ELSE 0 END) AS name_missing,
     SUM( CASE WHEN surname IS NULL THEN 1 ELSE 0 END) AS surname_missing,
     SUM( CASE WHEN email IS NULL THEN 1 ELSE 0 END) AS email_missing,
     SUM( CASE WHEN gender IS NULL THEN 1 ELSE 0 END) AS gender_missing,
     SUM( CASE WHEN race IS NULL THEN 1 ELSE 0 END) AS race_missing,
     SUM( CASE WHEN age IS NULL THEN 1 ELSE 0 END) AS age_missing,
     SUM( CASE WHEN province IS NULL THEN 1 ELSE 0 END) AS province_missing,
     SUM( CASE WHEN social_media_handle IS NULL THEN 1 ELSE 0 END) AS social_media_handle_missing,
     SUM( CASE WHEN channel2 IS NULL THEN 1 ELSE 0 END) AS channel2_missing,
     SUM( CASE WHEN recorddate2 IS NULL THEN 1 ELSE 0 END) AS recorddate2_missing,
     SUM( CASE WHEN duration_2 IS NULL THEN 1 ELSE 0 END) AS duration_2_missing,
FROM broadcast.public.tv;
```

   Returned output:

	# USERID_ID_MISSING	# NAME_MISSING	# SURNAME_MISSING	# EMAIL_MISSING	# GENDER_MISSING	# RACE_MISSING
1	0	0	0	0	0	0

   There are no missing values.

4. Checking for duplicates

```
--Checking for duplicates

SELECT userid, name, surname, email,
       COUNT(*) AS duplicate_count
FROM broadcast.public.tv
GROUP BY  userid, name, surname, email
HAVING COUNT(*) > 1;
```

5. Checking the data types in the Table

```
--Checking the data type of columns in the table

DESC TABLE tv;
```

6. Nnnn

```
--Change the data type of column reccorddate2 (its saved as a varchar)

SELECT userid,
       name,
       surname,
       age,
       gender,
       race,
       province,
       channel2,

       --time change
       TO_DATE(recorddate2, 'YYYY/MM/DD HH24:MI') AS clean_date,
       DAYNAME(clean_date) AS Day_of_week,
       TO_TIME(recorddate2, 'YYYY/MM/DD HH24:MI') AS time_only,
       TO_CHAR(time_only, 'YYYY/MM/DD HH24:MI') AS time_with_am_pm,
```

7. Age bucket

```
         --Age bucket
       CASE
           WHEN age BETWEEN 0 AND 12 THEN '0-12 Child'
           WHEN age BETWEEN 13 AND 17 THEN '13-17 Teenager'
           WHEN age BETWEEN 18 AND 24 THEN '18-24 Young Adult'
           WHEN age BETWEEN 25 AND 59 THEN '25-49 Adult'
          ELSE 'Older Adult'
       END AS age_bucket,
```

Created an age bracket to group individuals into meaningful, manageable categories for analysis and decision-making. Specifically, here's how I created an age bracket with differences like 12 years between categories:

Age bracket	Category
0-12	Child
13-17	Teenager
18-24	Young Adult
25-49	Adult
50+	Older

8.Time Bucket

```
        --time bucket
    CASE
        WHEN DATE_PART('hour', TO_TIMESTAMP(recorddate2, 'YYYY/MM/DD HH24:MI')) BETWEEN 0 AND 6 THEN 'Night'
        WHEN DATE_PART('hour', TO_TIMESTAMP(recorddate2, 'YYYY/MM/DD HH24:MI')) BETWEEN 7 AND 12 THEN 'Morning'
        WHEN DATE_PART('hour', TO_TIMESTAMP(recorddate2, 'YYYY/MM/DD HH24:MI')) BETWEEN 13 AND 18 THEN 'Afternoon'
        WHEN DATE_PART('hour', TO_TIMESTAMP(recorddate2, 'YYYY/MM/DD HH24:MI')) BETWEEN 19 AND 23 THEN 'Evening'
       ELSE 'Unknown'
   END AS time_bucket
FROM broadcast.public.tv;
```

Created a time interval to group individuals into meaningful, manageable categories for analysis and decision-making. Specifically, here's how I created an time interval:

Age bracket	Category
00:00 – 06:59	Night
07:00 – 12:59	Morning
13:00 – 18:59	Afternoon
19:00 – 23:59	Evening

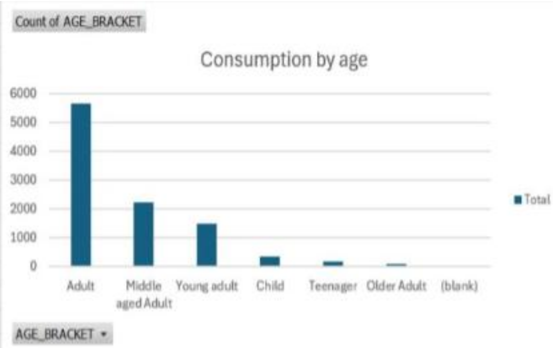
Trend Analysis

User Demographics:

1. Consumption By Age

   Downloaded document created from the age bracket join on snowflake for consumption by age into MS Excel to create a graph. Sorted the graph in descending order by count of age bracket to display the age group with the highest to lowest consumption.

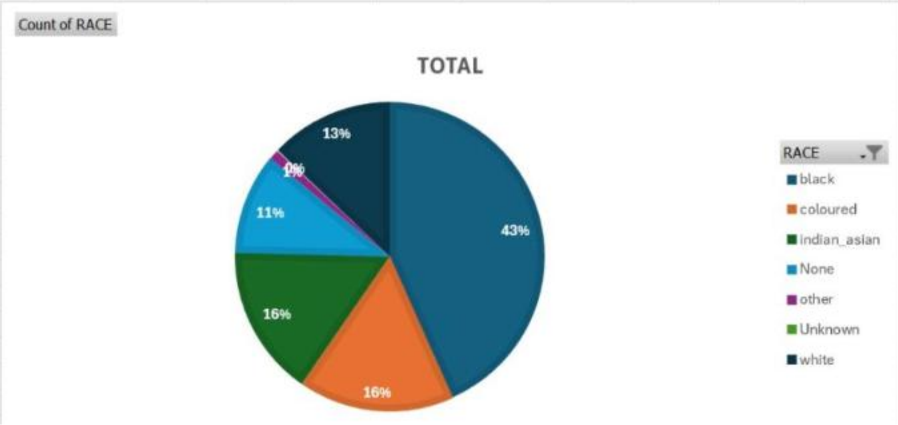
Row Labels	Count of AGE_BRACKET
Adult	5657
Middle aged Adult	2216
Young adult	1484
Child	359
Teenager	180
Older Adult	104
(blank)	
Grand Total	**10000**



2. Consumption by race

   Downloaded document created from the join on snowflake for consumption by race into MS Excel to create a graph. Made use of the pie chart to effectively display all the races and percentage of consumption

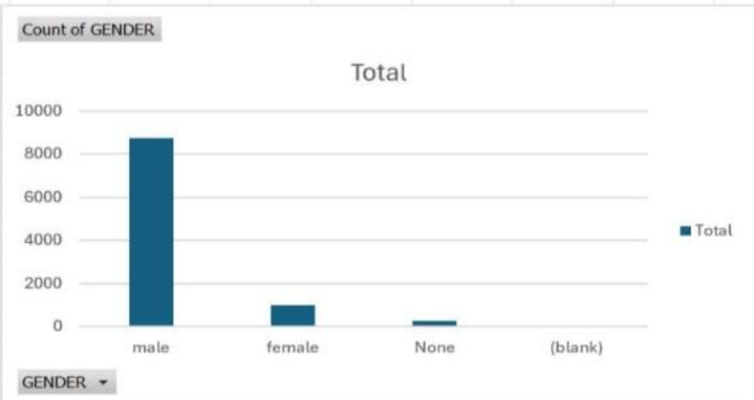
Row Labels	Count of RACE
black	4331
coloured	1633
indian_asian	1575
None	1057
other	102
Unknown	10
white	1292
Grand Total	**10000**



3. Consumption By Gender

Downloaded document created from the join on snowflake for consumption by race into MS Excel to create a graph. Sorted the graph in descending order by count of gender to display the gender with the highest to lowest consumption.

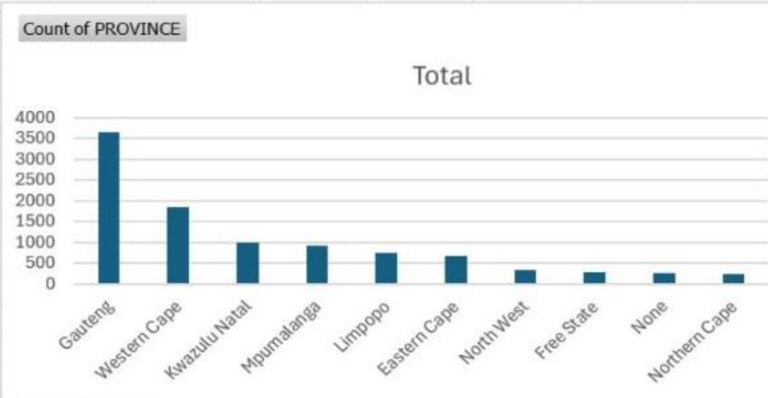
Row Labels	Count of GENDER
male	8761
female	977
None	262
(blank)	
Grand Total	**10000**



4. Consumption By location.

Downloaded document created from the join on snowflake for consumption by race into excel to create a graph. Sorted the graph in descending order by count of location to display the location with the highest to lowest consumption.

Row Labels	Count of PROVINCE
Gauteng	3654
Western Cape	1845
Kwazulu Natal	1001
Mpumalanga	918
Limpopo	763
Eastern Cape	690
North West	344
Free State	292
None	263
Northern Cape	230
Grand Total	**10000**

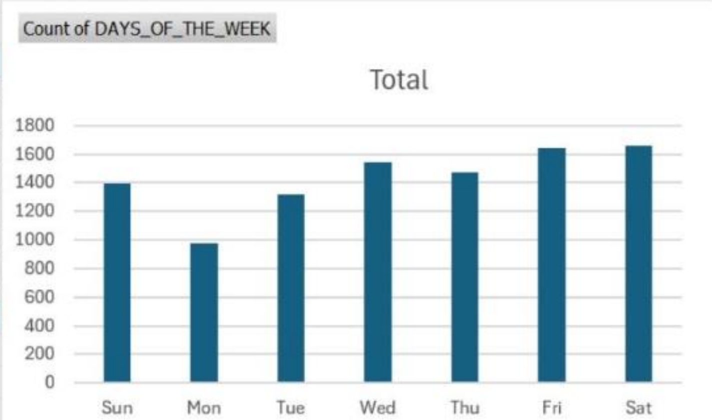


User Trends:

1. Peak Consumption days

   Downloaded document created from the inner join on snowflake for consumption by race into excel to create a graph. Sorted the graph in descending order by count of days of the week to display the days of the week with the highest to lowest consumption.

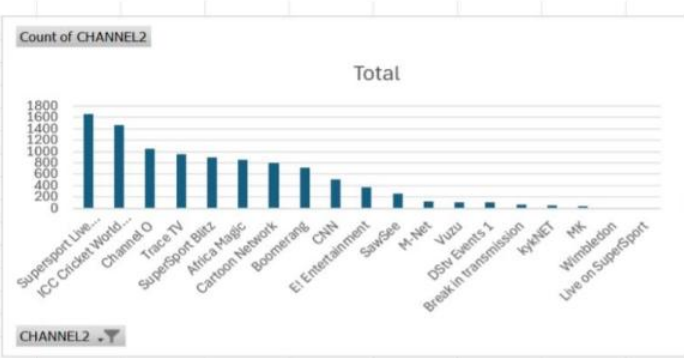
Row Labels	Count of DAYS_OF_THE_WEEK
Sun	1398
Mon	974
Tue	1321
Wed	1539
Thu	1471
Fri	1642
Sat	1655
Grand Total	**10000**



2. Top 5 Most watched Channels

   Downloaded document created from the inner join on snowflake for consumption by race into MS Excel to create a graph. Sorted the graph in descending order by count of channel to display the channel with the highest to lowest consumption. From this bar graph I was able to create a table with the top 5 most watched channels with number of views and percentage of total viewership.

Row Labels	Count of CHANNEL2
Supersport Live Events	1662
ICC Cricket World Cup 2011	1465
Channel O	1050
Trace TV	952
SuperSport Blitz	896
Africa Magic	859
Cartoon Network	793
Boomerang	714
CNN	505
E! Entertainment	367
SawSee	255
M-Net	116
Vuzu	111
DStv Events 1	107
Break in transmission	66
kykNET	45
MK	32
Wimbledon	3

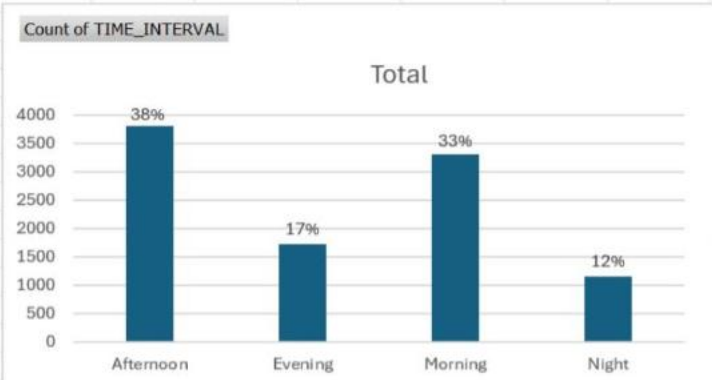


Top 5 Channels	Views	Percentage
1.SuperSport Live Events	1662	17%
2.ICC Cricket world Cup 2011	1465	15%
3. Channel O	1050	11%
4. Trace TV	952	10%
5.SuperSport Blits	896	9%

3. Peak consumption days

Downloaded document created from the time bracket on snowflake for consumption by time only into MS Excel to create a graph. From this bar graph I was able to create a table with the top 5 peak consumption times.

Row Labels	Count of TIME_INTERVAL
Afternoon	3809
Evening	1730
Morning	3304
Night	1157
Grand Total	**10000**



Insights drawn from Trend Analysis

From the users demographics I can conclude that BrightTvs main consumers are black men that range from the ages of 18-39 who reside in Gauteng.
From the users trends I can conclude that BrightTvs main consumers days are Saturday and Friday with our top 5 channels falling under music and sports channels.

Factors Influencing Consumption

- Sports events draw massive viewership, particularly live events.
- Music channels attract consistent engagement.
- Prime time evenings, weekends(Friday and Saturday) and public holidays are high-consumption periods.
- Group activities such as family viewings or sports gatherings likely amplify engagement.

Recommendations to increase consumption on low consumption days

- Run promotional campaigns targeting Gauteng residents using ads highlighting popular content like sports and music.
- Partner with local influencers, particularly those in sports and entertainment industries.
- Offer “refer-a-friend” discounts or free trial subscription(1 month free)
- Partner with sports leagues(e.g. cricket or football) to offer live-streaming of events exclusively.
- Collaborate with artists featured on music channels (e.g. Channel O, Trace TV) for exclusive content.
- Organize viewing events for big sports matches or concerts in local venues. Collaborate with TikTok influencers for visibility and engagement with their audiences.
- Expand BrightTvs reach by leveraging partnership with mobile carries or bundling services with internet providers.

Initiative to grow BrightTvs user base

- Introduce weekday- specific programming (Monday Sports Recap or Midweek movie night)
- For Sunday: Cultural or inspirational programming(e.g. African Storytelling, religious content)
- Allow viewers to vote for favourite shows or curate playlist, particularly on low-consumption days.
- Replay iconic matches or moments from SuperSport Live Event to engage sports fans even during off-peak days.